Lab - Blob Snapshots and Versioning
To Begin with the Lab

Summary

This lab demonstrates how to protect blob data using **Blob Snapshots** and **Blob Versioning** in Azure Storage. Snapshots create read-only copies of a blob at a specific point in time, while Versioning automatically keeps previous versions whenever a blob is modified or deleted, making it easy to recover data.

Understanding

Azure Blob Storage provides two important data protection features:

- **Blob Snapshots** create a **read-only copy** of a blob at a specific point in time. Snapshots are created manually and remain available until they are deleted. They are useful before making changes to important files because they allow you to restore the original content if needed.
- **Blob Versioning** automatically creates a **new version** every time a blob is modified or overwritten. Each version has a unique version ID, allowing you to view, restore, or download previous versions without manually creating snapshots.

Using snapshots and versioning together helps prevent accidental data loss, enables recovery from unwanted changes, and supports backup and auditing requirements.

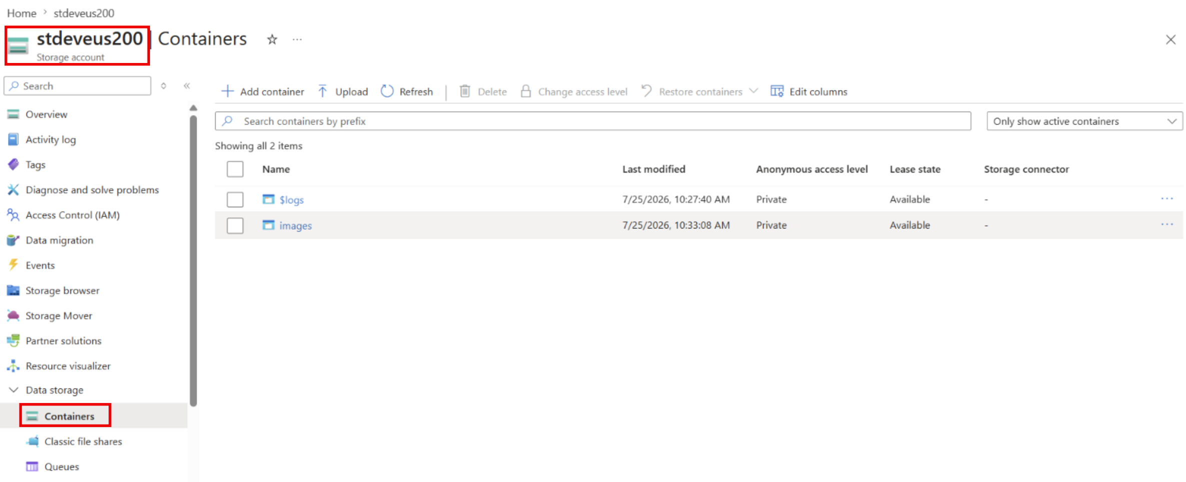
Example:

Suppose a document named **report.pdf** is stored in Azure Blob Storage. Before editing it, you create a snapshot. Later, another user updates the file. Since Blob Versioning is enabled, Azure automatically saves the previous version. If the latest version contains incorrect information, you can restore either the snapshot or a previous version.

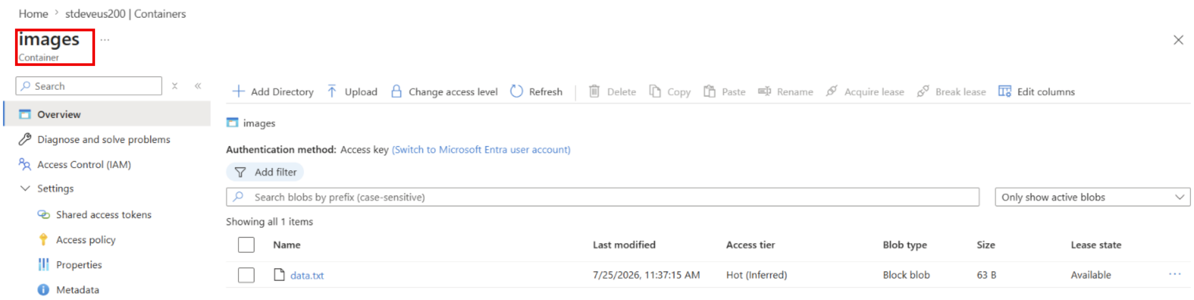
Lab Steps

Step 1: Open the Storage Account

- Sign in to the Azure Portal.
- Open your **Storage Account**.
- Navigate to **Containers**.

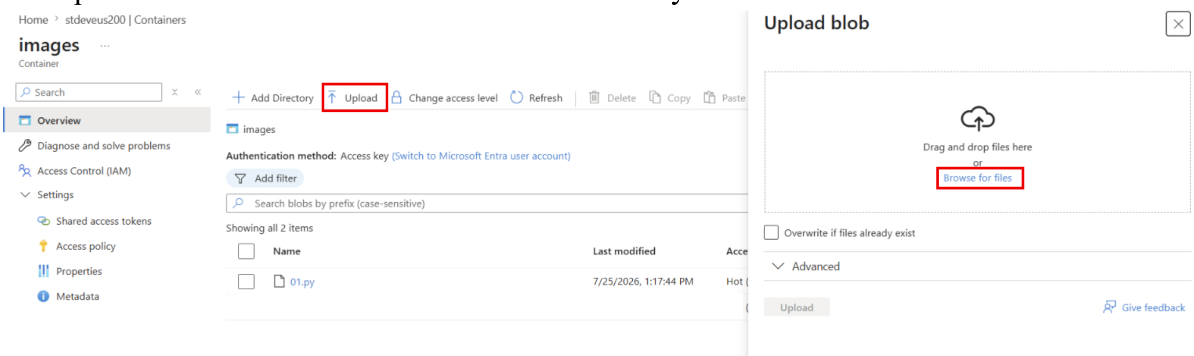


- Open the **container(images)** containing the blob.

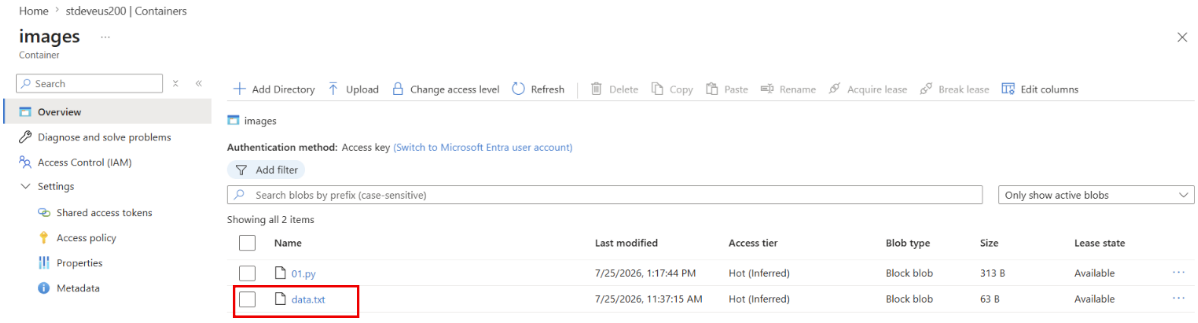


Step 2: Upload a Blob

- Click on the **Upload**
- Upload a file to the container if one is not already available.

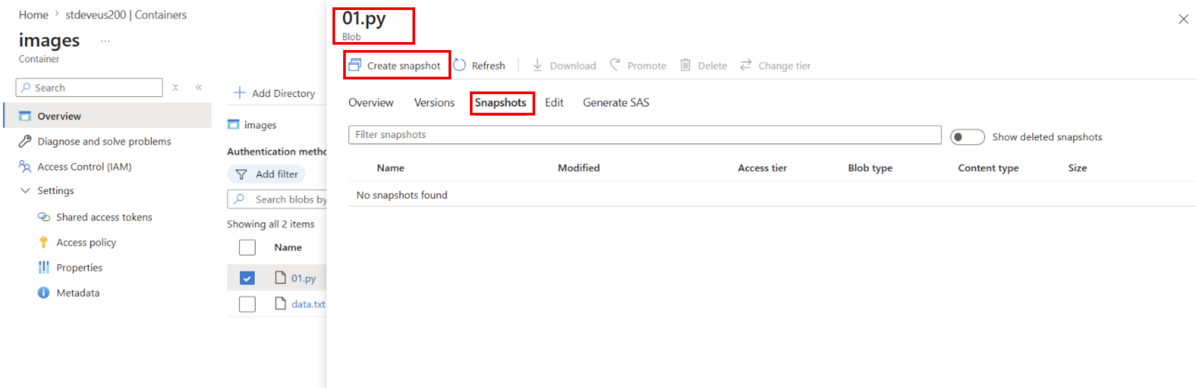


- Verify that the blob appears in the container.

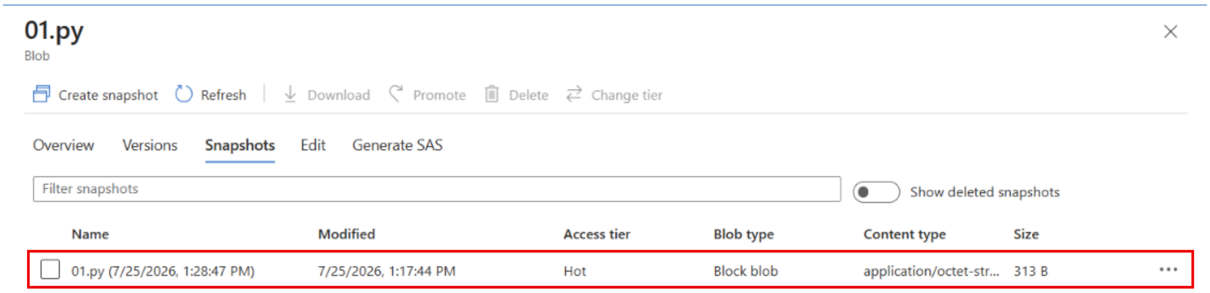


Step 3: Create a Blob Snapshot

- Select the blob.
- Navigate to **Snapshot**.
- Click **Create Snapshot**.

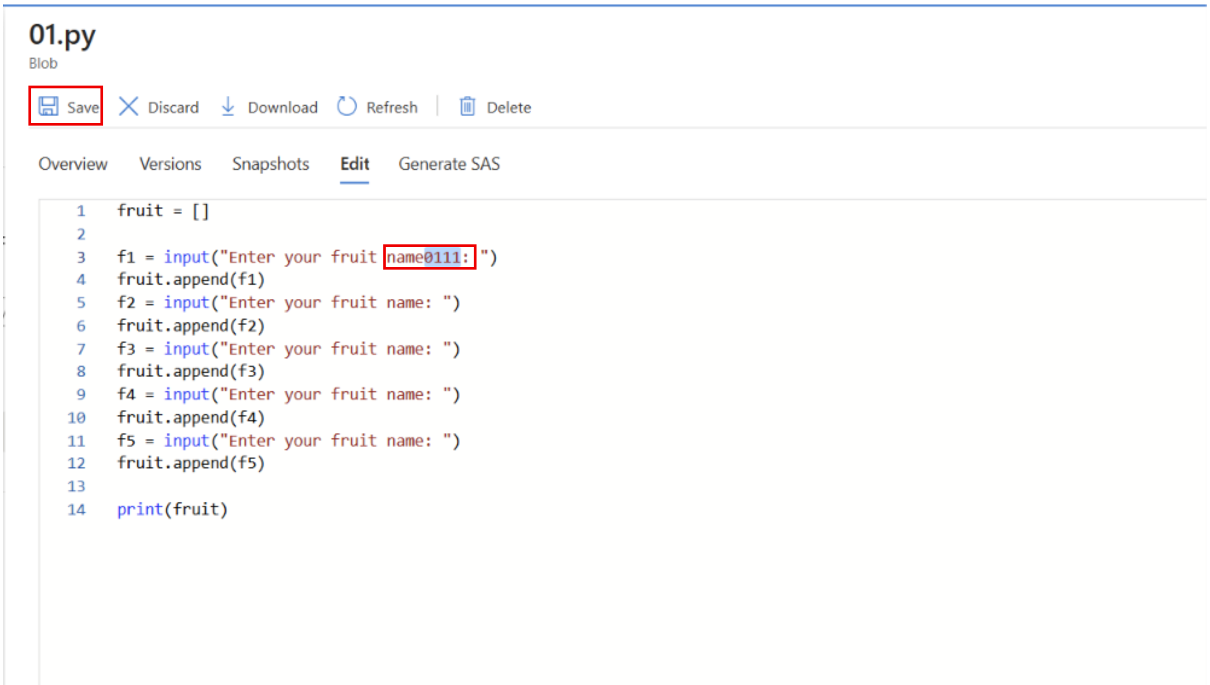


- Azure creates a read-only snapshot of the blob.
- Verify that the snapshot appears under the blob.

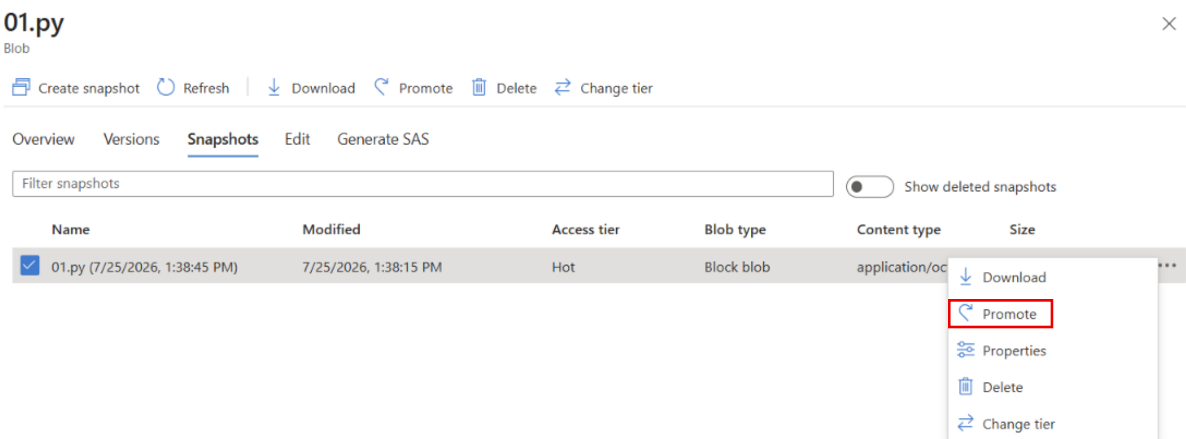


Step 4: Modify the Blob

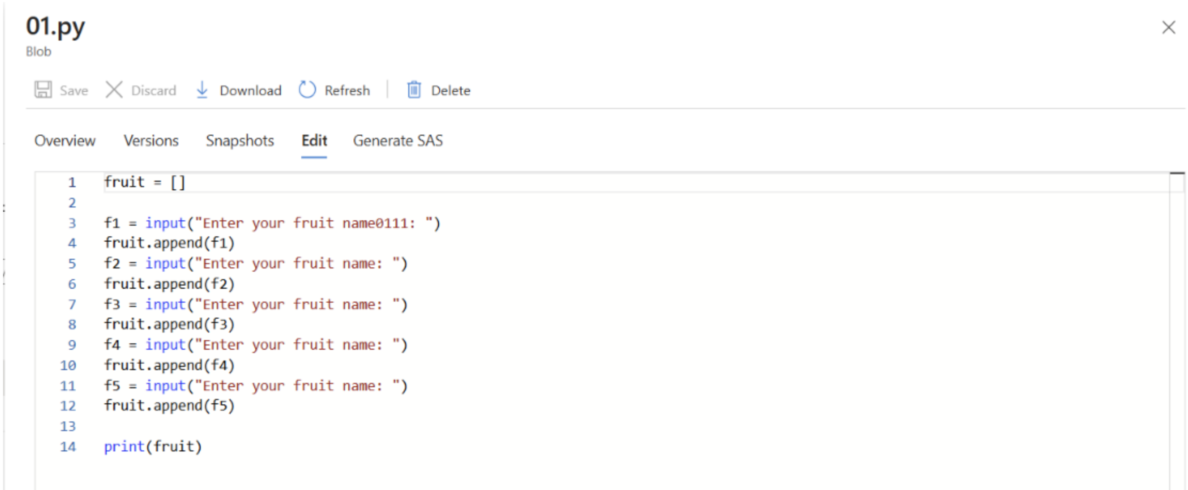
- Upload a new version of the same file.



- Choose **Overwrite** when prompted.

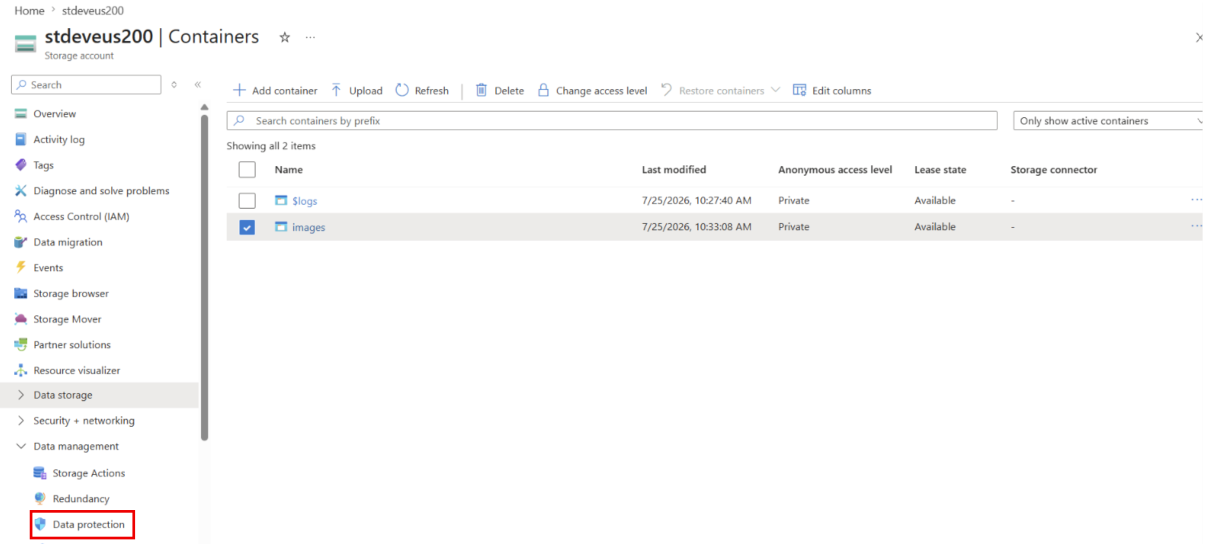


- Verify that the blob content has changed.

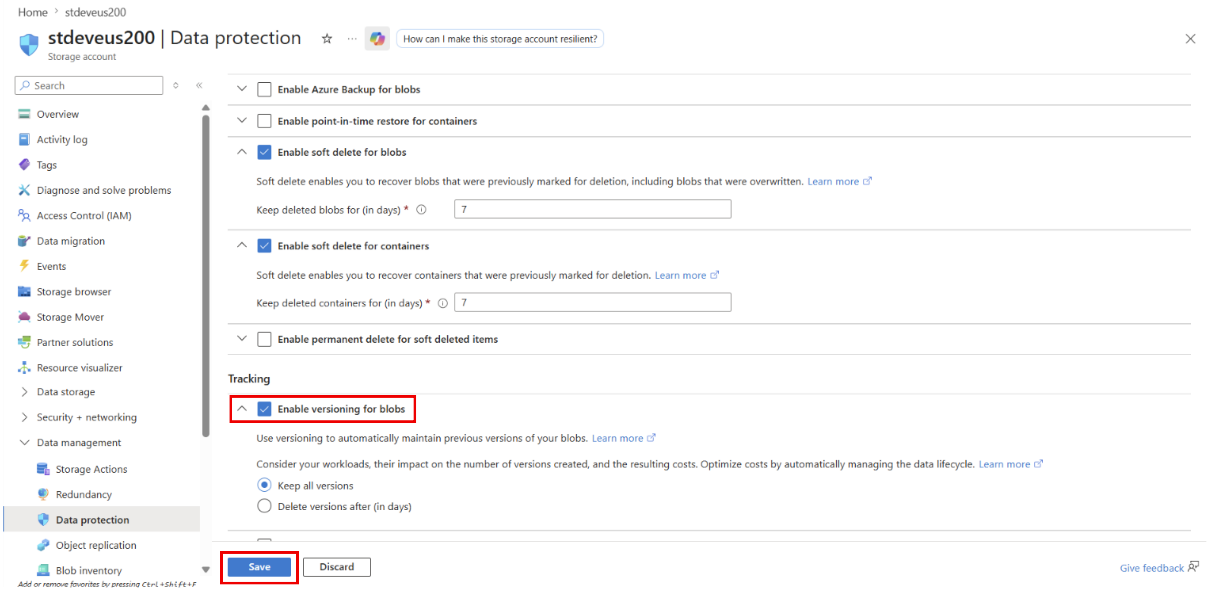


Step 5: Enable Blob Versioning

- Return to the **Storage Account**.
- Open **Data Protection**.

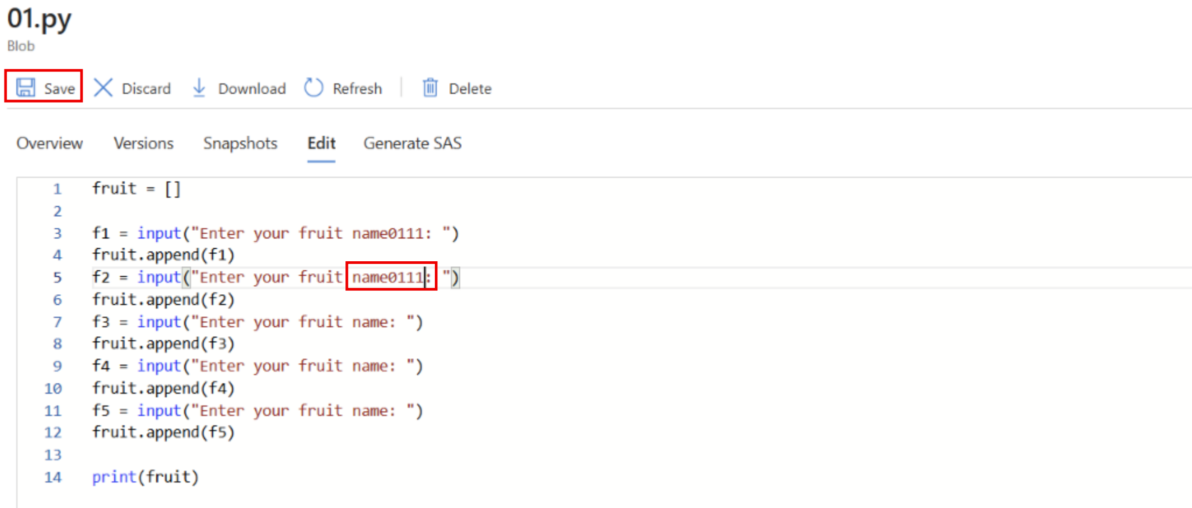


- Enable **Blob Versioning**.
- Save the changes.



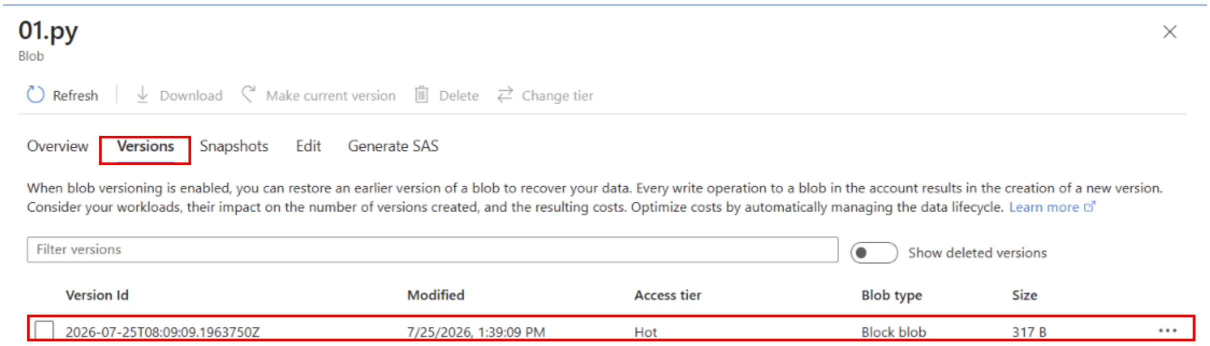
Step 6: Upload a New Version

- Return to the container.
- Upload the same file again with updated content.
- Overwrite the existing blob.
- Azure automatically creates a new blob version.



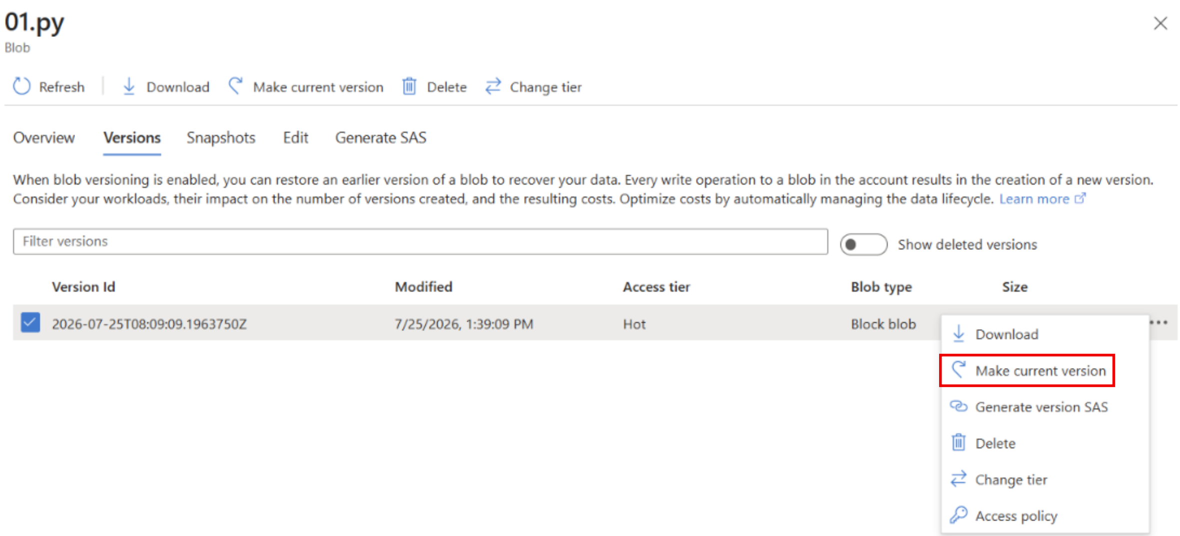
Step 7: View Blob Versions

- Select the blob.
- View the list of available versions.
- Observe that each version has a unique Version ID and creation time.

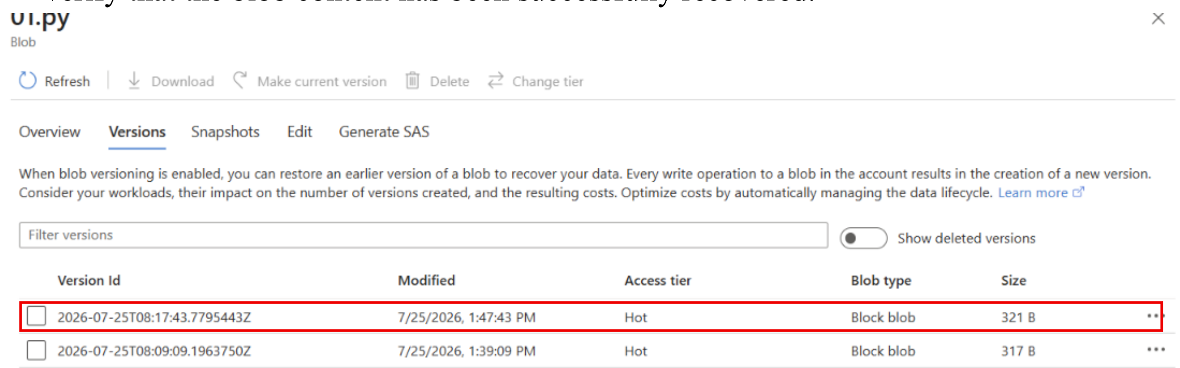


Step 8: Restore a Previous Version

- Select an older version of the blob.
- Restore or promote it as the current version.

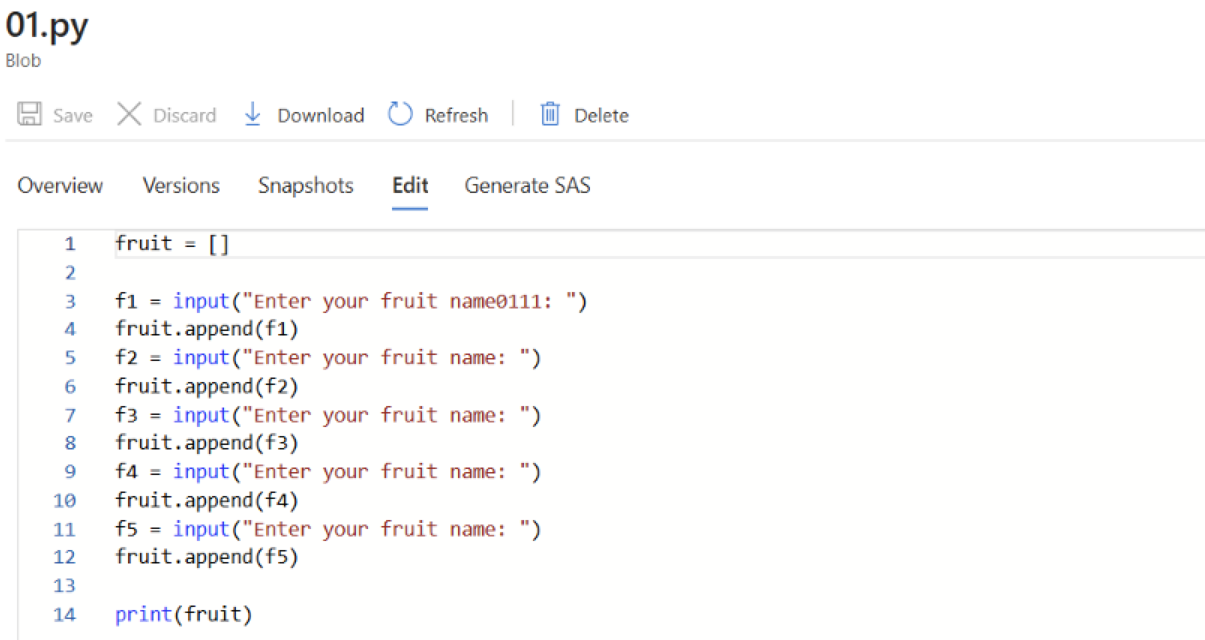


- Verify that the blob content has been successfully recovered.



Step 9: Access Blob Snapshots

- View the snapshots associated with the blob.
- Compare the snapshot with the current blob.



Step 10: Verify Data Recovery

- Confirm that both snapshots and versions can be used to recover previous data.

- Ensure the correct version is available after restoration.